

It hopes to make it so easy that electronics can be woven into everyday garments! Their solution is based on conductive yarns, which combine thin, metallic alloys with natural and synthetic yarns like cotton, polyester or silk. The resulting Jacquard yarns are just like traditional yarns, and can be woven on any industrial loom. These yarns can be attached to discreet connectors and miniaturised circuits, to weave different kinds of technological applications into the dress. Some areas, for example, can be touch- and gesture-sensitive, while others can be sensor grids. Collected data can be wirelessly transmitted to mobile devices to provide varied services to the user.

The ability to manufacture the yarns and fabrics on standard looms used by mills across the world is, according to Google, the biggest benefit of this method. They have partnered with Levi's to create interactive garments, which combine the traditional charm of denim with the benefits of technology.

One such product is Levi's Commuter Trucker Jacket for urban bike commuters. When they are driving and cannot access their smartphones, wearers can control their mobile experience and connect to a variety of services such as music or maps, directly from the jacket.

A plastyck solution. At K 2016 plastics trade fair, Covestro, one of the world's leading polymer companies showcased luminous clothing made with freely-formable electronic systems, which are responsive to movements without losing their functionality. The garments use a flexible and formable film made of thermoplastic polyurethane (TPU) from Covestro.

TPU is the substrate for printed copper circuits, which are arranged in a meandering pattern and can, thus, also be bent and stretched. This technology is based on stretchable circuit board (SCB) technology developed



Fig. 2: Owllet Smart Sock is a comfortable way to monitor babies (Image courtesy: Owllet)

earlier as part of European STELLA project, led by Fraunhofer Institute for Reliability and Microintegration (IZM) and Technical University of Berlin.

According to the press release, the smart circuits are manufactured using an efficient, multi-stage process. The release states, "First, copper films are laminated onto polyurethane films. Printed circuits are produced in a subsequent structuring operation featuring highly effective adhesion. Coated films are then shaped as required using conventional thermoforming. Formable electronic systems can be directly laminated into textiles."

Pretty and purposeful embroidery. A team at The Ohio State University is developing a way of embroidering antennae and circuits into fabrics, with 0.1mm precision—a size ideal for integrating components like sensors and computer memory devices into clothing. Their method can be implemented using a typical tabletop sewing machine, except that it embroiders its patterns with fine silver metal wires. The thread has a diameter of 0.1mm and is made of seven filaments. Each filament has a core of copper, encased with pure silver.

The shape of the embroidery determines its functionality, say, frequency of operation in the case of an antenna. A broadband antenna, for example, comprises more than six small interlocking geometric shapes that form an intricate circle a few centimetres wide. Each piece

of the circle transmits energy at a different frequency, so that these cover a broad spectrum of energies when working together, thereby enabling broadband capability for mobile phone and Internet access.

Other shapes serve other functions. There is a spiral-shaped antenna that helps improve mobile phone signal reception. Measuring 15.2cm (6-inch) across, this antenna can

transmit signals at frequencies of 1GHz to 5GHz with near-perfect efficiency. Embroidered on the back of your dress, this spiral pattern could help boost the reception of your mobile device.

Screen printing tech onto fabrics.

Screen printing is another common method used to print designs on fabrics. Germany's Fraunhofer Institute for Silicate Research hopes to use this process to bring tech into textiles. They have developed piezoelectric polymer sensor printing pastes that can be applied to fabrics using a simple screen printing process. After printing, the sensors are subjected to an electric field, making the piezoelectric polymers align to adopt the targeted pressure sensitivity.

The ferroelectric polymer paste registers pressure and deformation, which means that it can act as a touch-and-motion sensor. It is also pyroelectric or sensitive to changes in temperature, and can be used as a proximity sensor. The printed sensors are non-toxic, thinner than human hair, transparent and flexible. The team claims that the biggest benefit of this technology is that it does not require a power source. It can run using energy harvested from the wearer's movements.

Powering up. To be embedded into fabrics, batteries need to be flexible and, preferably, stretchable. Inspired by a Japanese paper-folding technique called Kirigami, a team from Arizona State University has developed a lithium-ion based battery that can be stretched to 150 per